MC2MC

The Dawn Of The Mesh; Zero-trust at scale

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About us

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What is Zero-Trust?

Explicit Verification

Least Privilege

Assume Breach

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The 7 tenets of Zero Trust according to NIST

1. All data sources and computing services are considered resources.
2. All communication is secured regardless of network location.
3. Access to individual enterprise resources is granted on a per-session basis.
4. Access to resources is determined by dynamic policy-including the observerable state of client identity, application/service, and the requesting asset – and may include other behavioral attributes.
5. The enterprise monitor and measures the integrity and security posture of all owned and associated assets.
6. All resource authentication and authorization are dynamic and strictly enforced before access is allowed.
7. The enterprise collects as much information as possible about the current state of assets, network infrastructure and communications and uses it to improve its security posture.

Source: https://nvlpubs.nist.gov/nistpubs/specialpublications/NIST.SP.800-207.pdf

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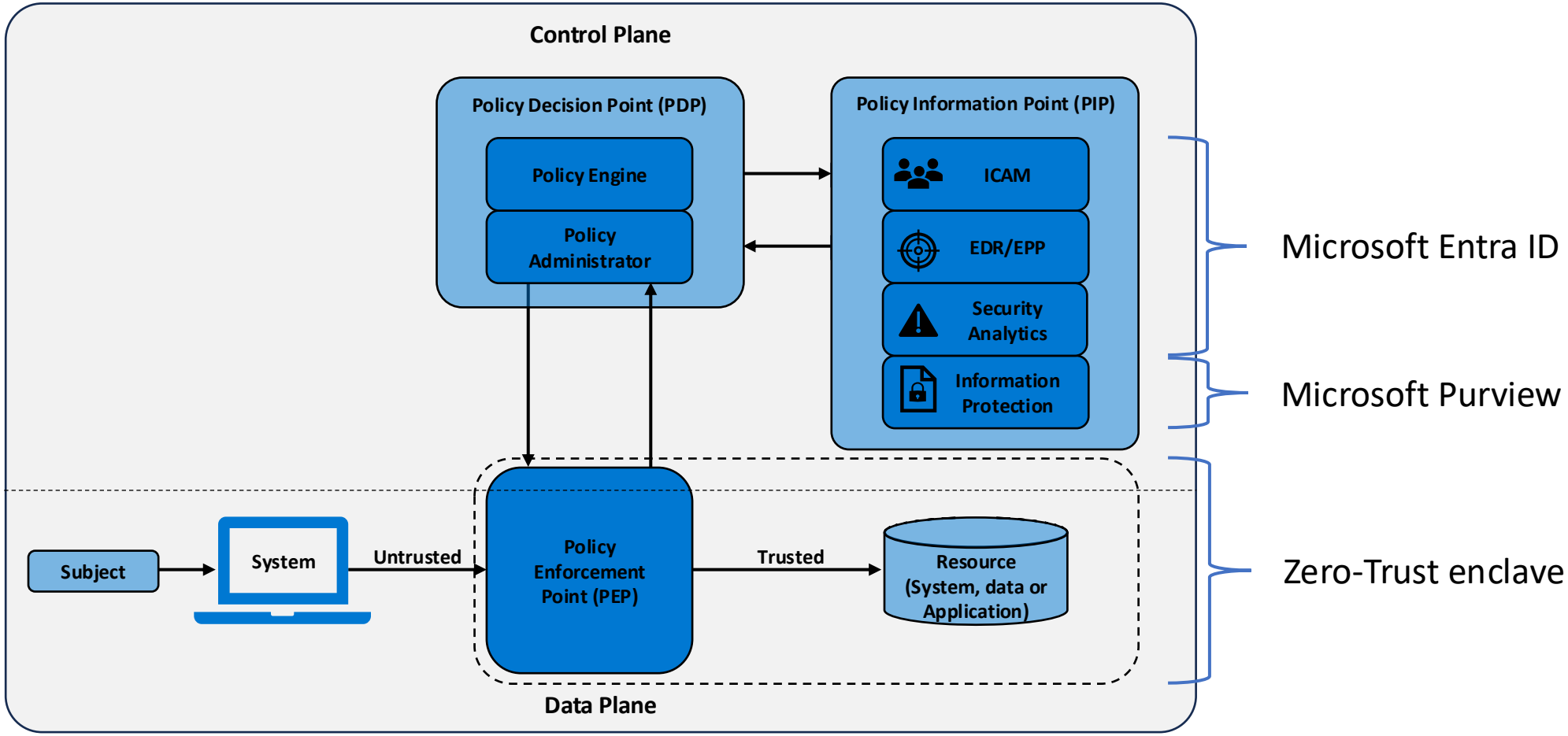
What does Zero-Trust look like?



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Identity

Identity is the cornerstone of Zero-Trust

- Access to individual enterprise resources is granted on a per-session basis.
- Access to resources is determined by dynamic policy-including the observerable state of client identity, application/service, and the requesting asset – and may include other behavioral attributes.
- All resource authentication and authorization are dynamic and strictly enforced before access is allowed.

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Azure Enterprise Landingzone

- Does this affect the Azure Enterprise Landingzone?!
- You do have to rely more on Policy Drive Governance
- Consider providing immutable building blocks.

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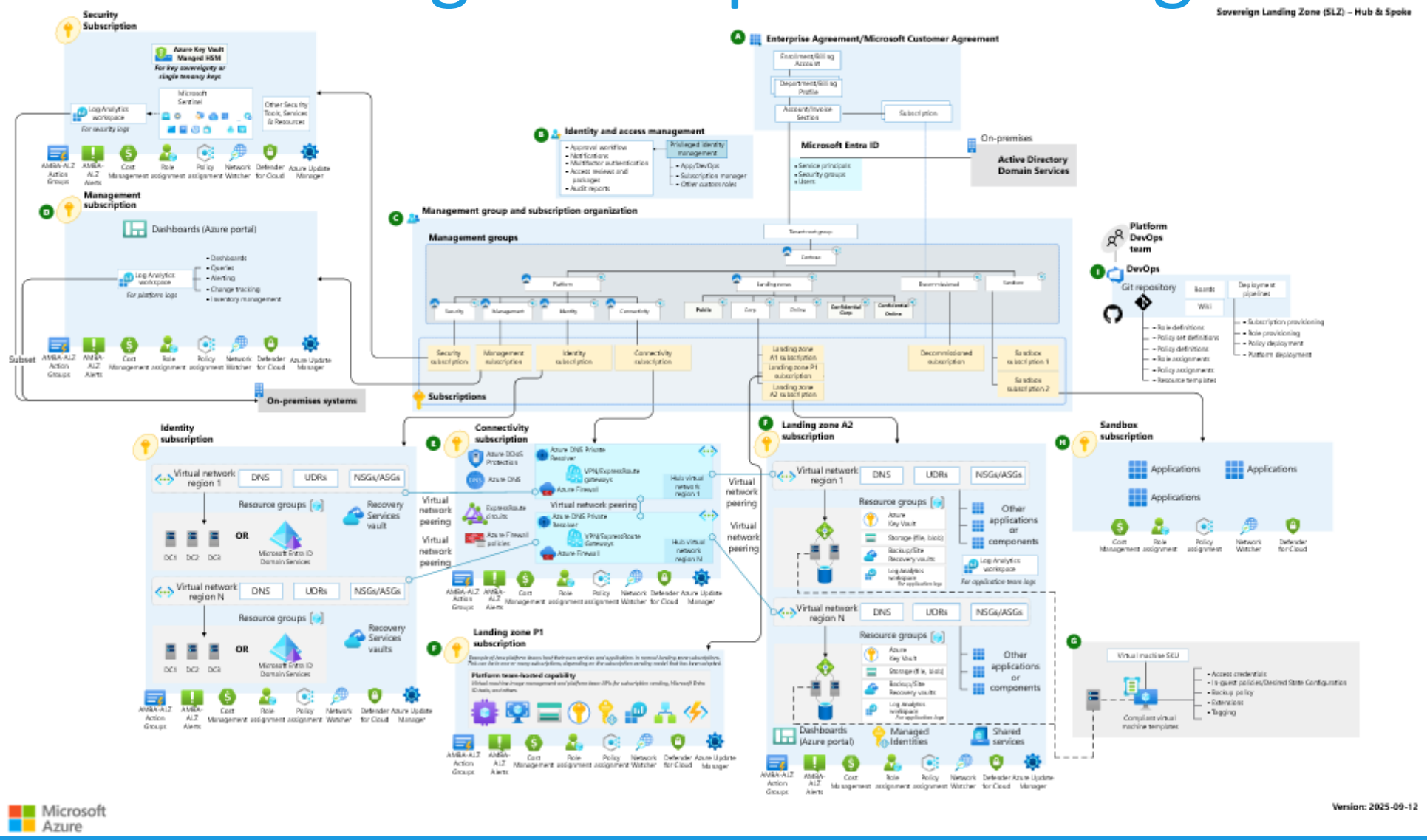
The Soevereign Enterprise Landing Zone



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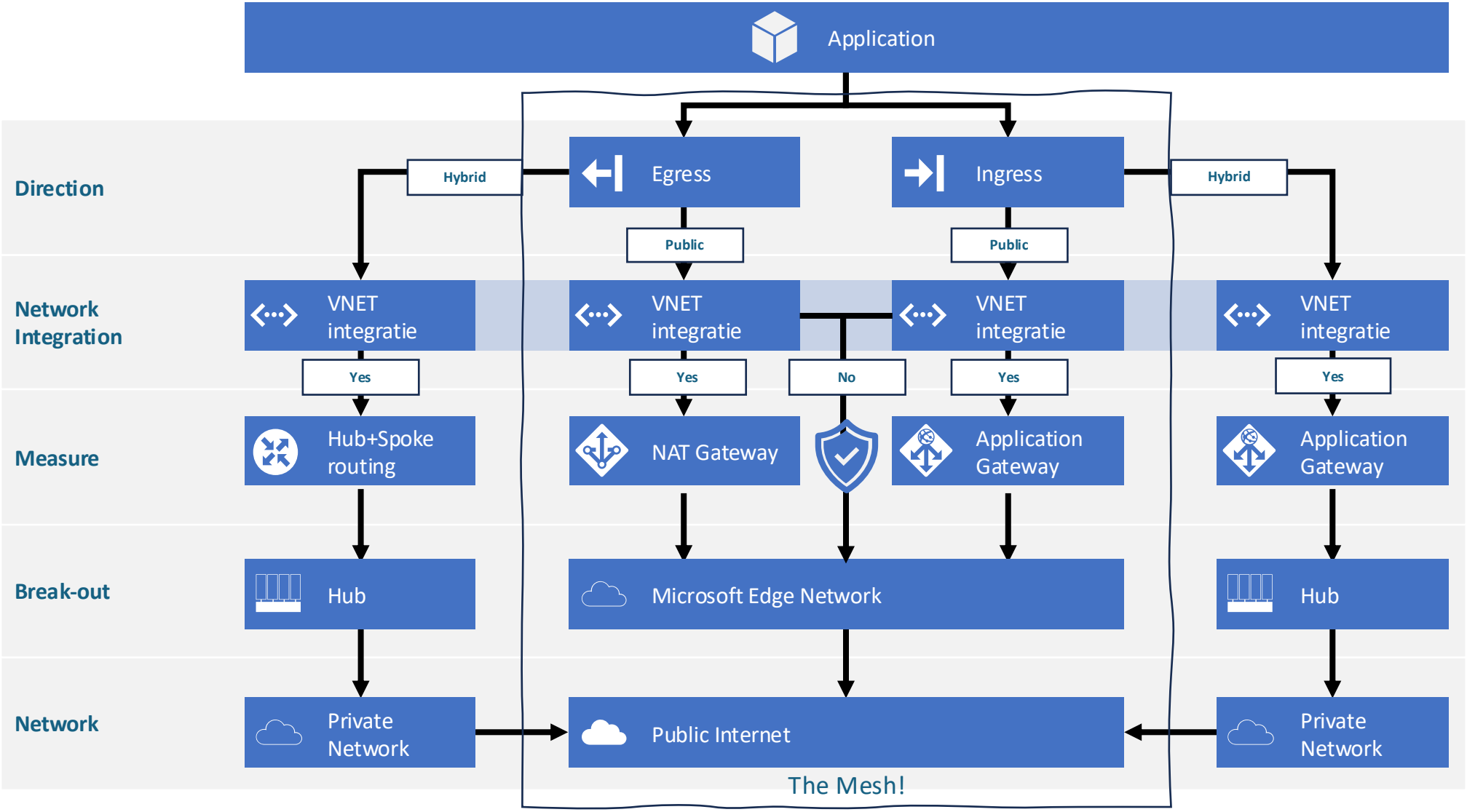
Connectivity patterns



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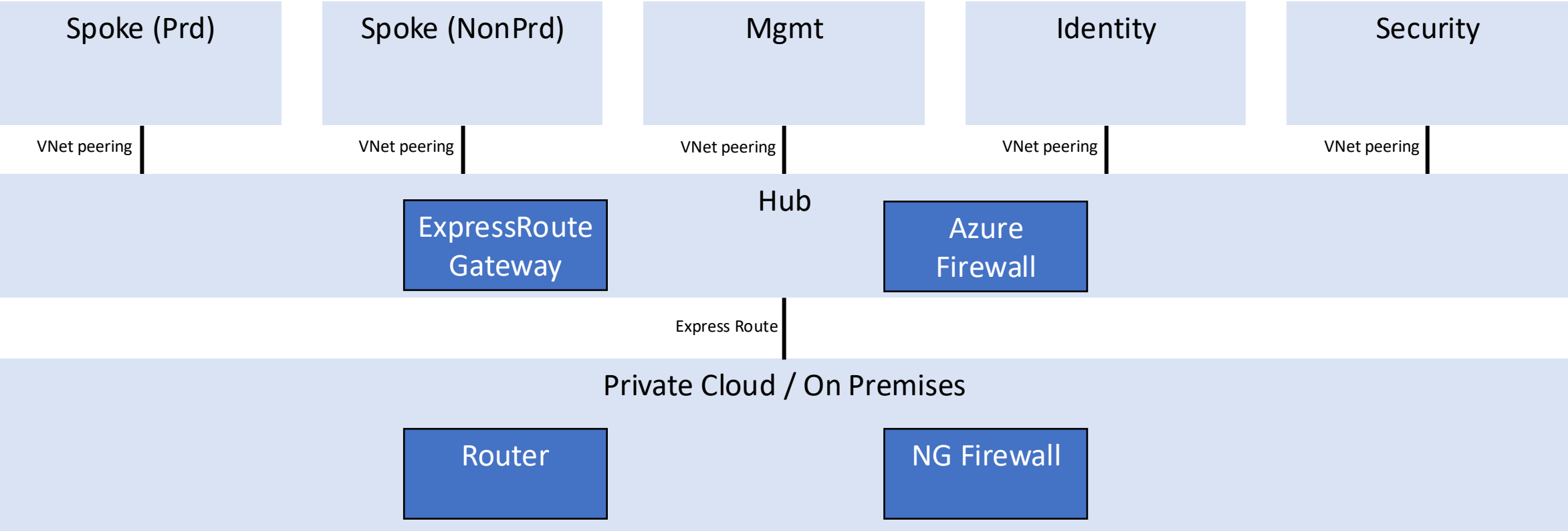
The traditional Hub+Spoke model



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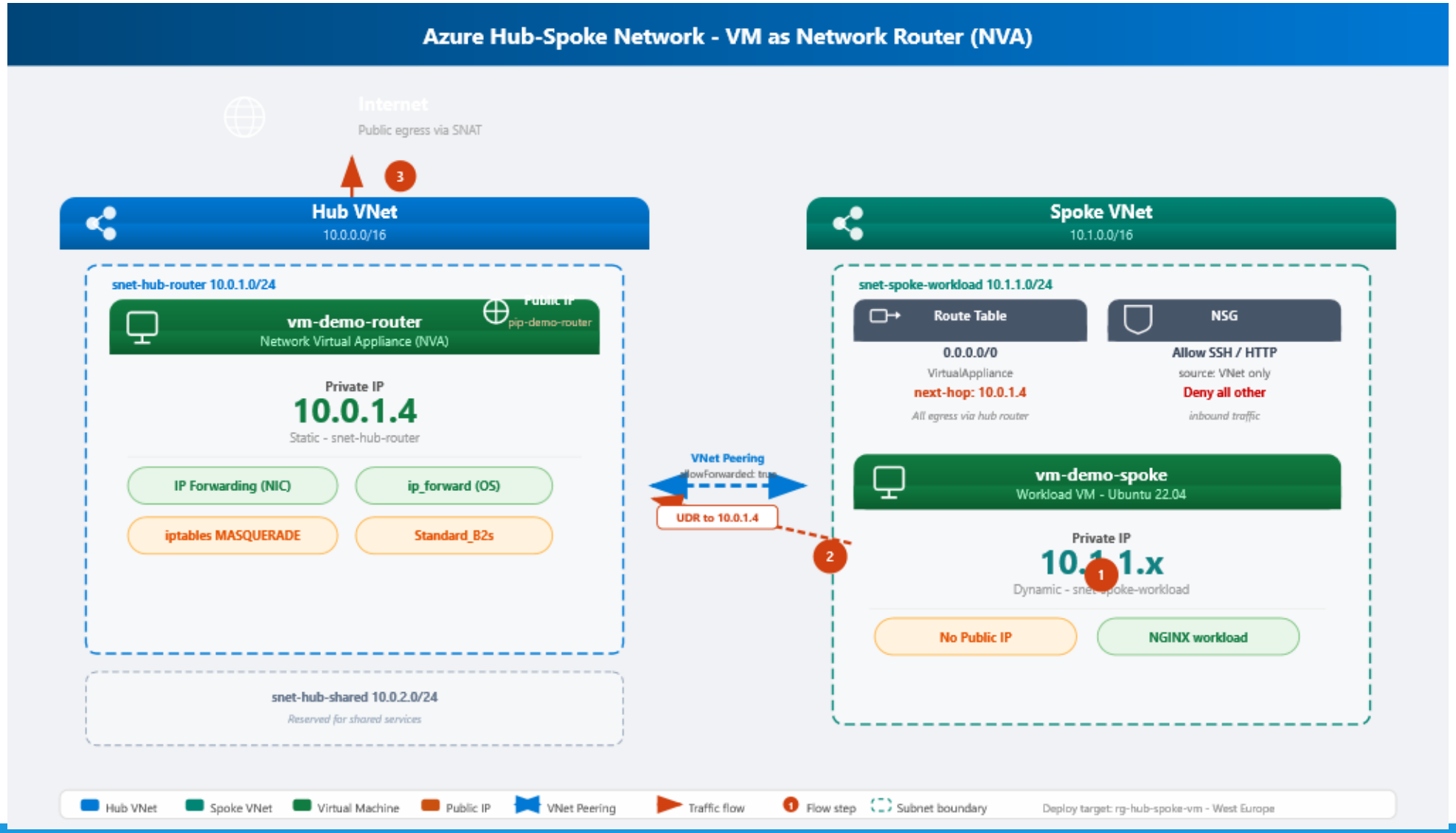
Azure demo Hub+Spoke



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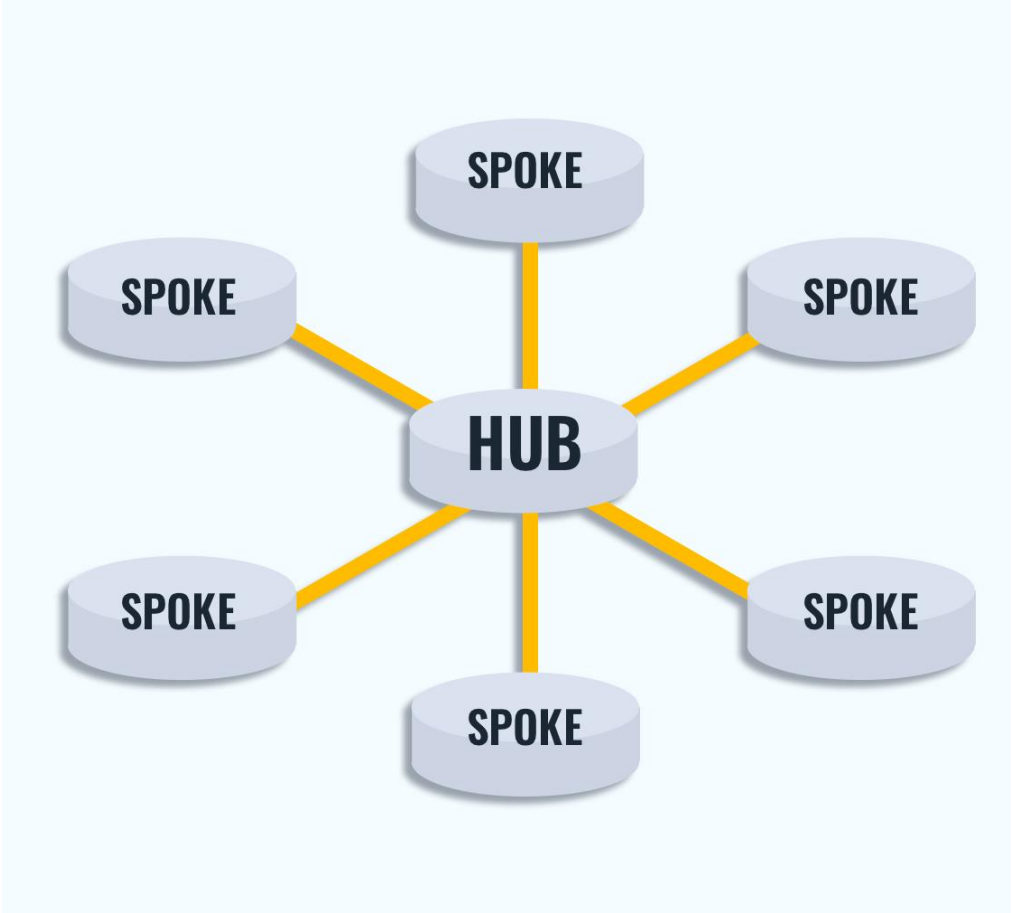
Azure demo Hub+Spoke



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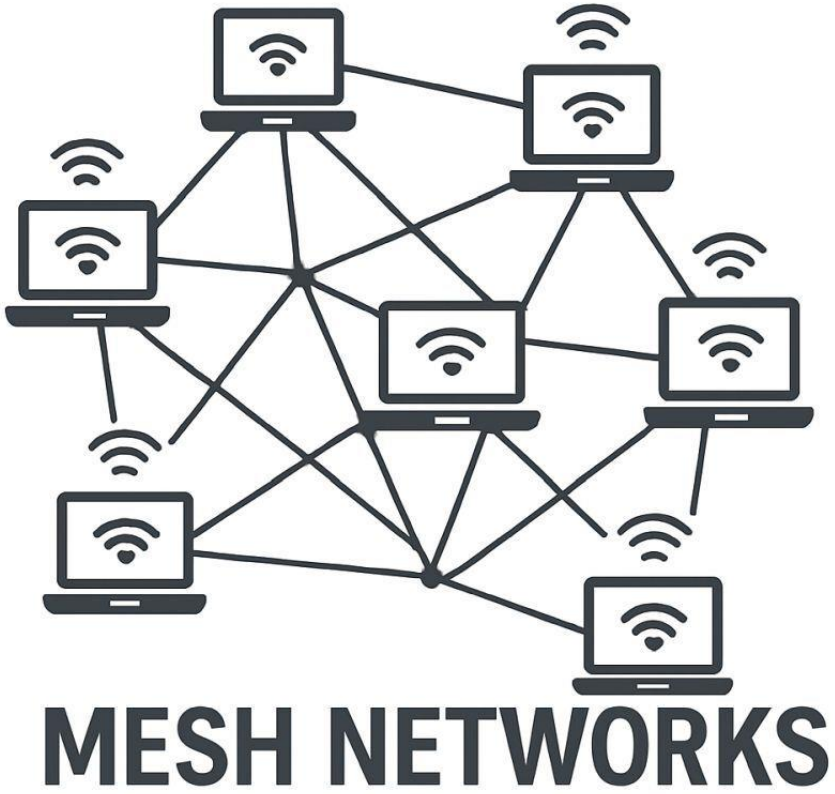
A Zero-Trust enclave in Azure and now….. It’s getting Meshy



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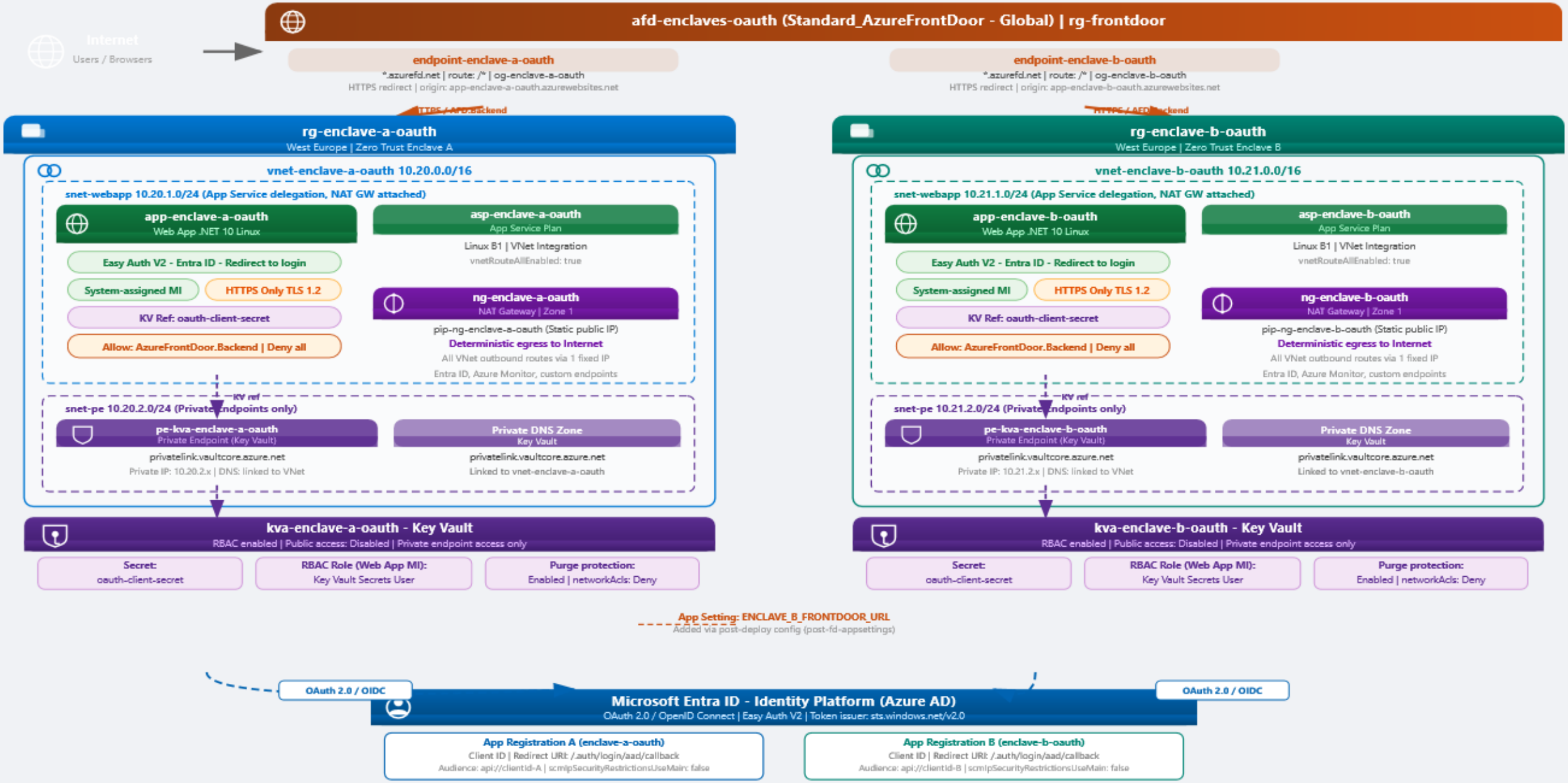
Zero Trust Enclaves with OAuth - Entra ID Easy Auth V2 / Azure Front Door Standard



Azure demo Mesh

- Let’s make a nice stroll through resource avenue

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Find the differences

- How long will that firewall change take?!
- Can’t we do that ourselves?!

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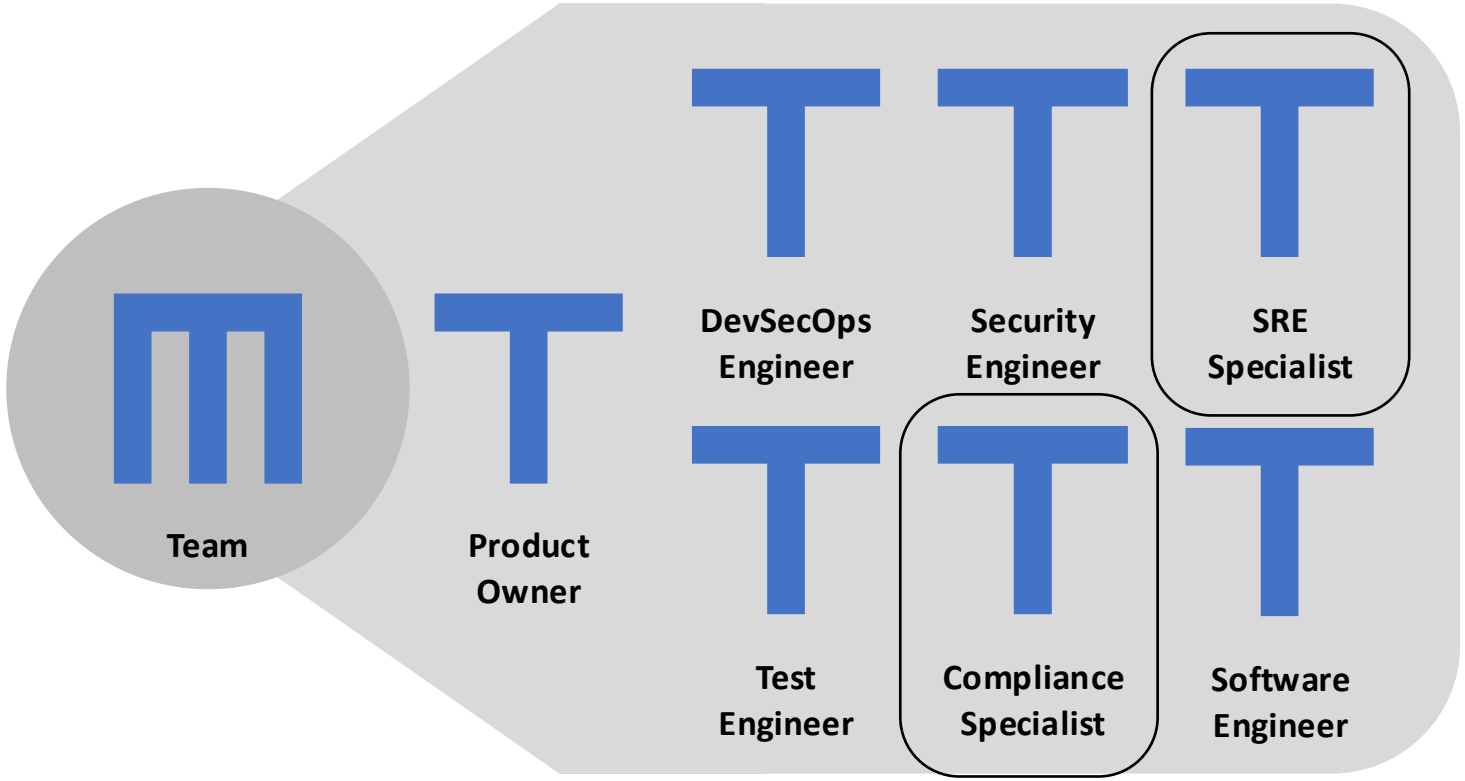
The anatomy of a DevSecOps team



Example view of a DevSecOps team

- A DevSecOps team is M-shaped consisting out of T-shaped professionals.
- These professionals are inhouse native or externally sourced
- There's a clear insourcing strategy

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Autonomy of DevSecOps teams

- Teams can do more themselves!
  - No changes outside your own team ☺
- Teams have to do more themselves!

"With great powers come great responsibilities!"

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Hub+Spoke vs Mesh (1/2)

Aspect	Hub + Spoke	Mesh
Zero-Trust implementation	+ Central inspection and policy enforcement in one place – Single chokepoint; harder to eliminate lateral movement	+ Policies close to workloads; fine-grained segmentation + No central bottleneck; easier to enforce least-privilege per service
DevSecOps autonomy	– Strong dependency on platform/network teams (IPAM, routing, firewall changes) – Slow change cycles	+ Teams manage their own connectivity via DNS/FQDNs + Faster iteration, fewer dependencies
Scalability	+ Works well for stable, predictable environments – Hub becomes a bottleneck when scaling or going multi-region	+ Horizontally scalable; no central choke point + Ideal for dynamic workloads and microservices
Complexity	+ Simple for central teams to understand and govern – Complexity shifts to routing and firewall management	+ Distributed complexity but highly automatable + Fits GitOps/Infrastructure-as-Code

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Hub+Spoke vs Mesh (2/2)

Aspect	Hub + Spoke	Mesh
Latency & performance	– Traffic flows through the hub → extra hops – Can become expensive due to central appliances	+ Direct workload-to-workload connections + Lower latency, better performance
IP management (IPAM)	– Heavy dependency on central IP management – Subnet changes have wide impact	+ Minimal IP dependency + Service discovery via DNS → fewer disruptive changes
Security architecture	+ Strong central control – Harder to apply workload-specific policies	+ Zero-Trust aligned: identity-based, workload-centric + Policies per workload/service
Cost	– Expensive central firewalls, gateways, appliances – More operational load on platform teams	+ Costs distributed across workloads + Fewer central appliances required
Organizational fit	– Fits traditional IT governance– Less suitable for autonomous product teams	+ Fits modern DevSecOps and platform engineering models + Teams can deliver faster

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Lessons learned

- It’s always DNS!
- Identity is key!
- Security monitoring is more important then ever!
- A DevSecOps team is truly mature!
  - There’s no DevOps team without Sec!
- The vending machine is more than just new subscriptions;
  - DevOps environment
  - Agent Pools
  - Everything is really as-code!!!!!
  - NO PRIVILEGED PORTAL ACCESS IN PROD! (Principle of Least Privilege)
- Secrets and key-management!
  - And you get a Keyvault, and you get a Keyvault, everyone get’s a Keyvault!
  - Still using secrets?
- Provide immutable building blocks

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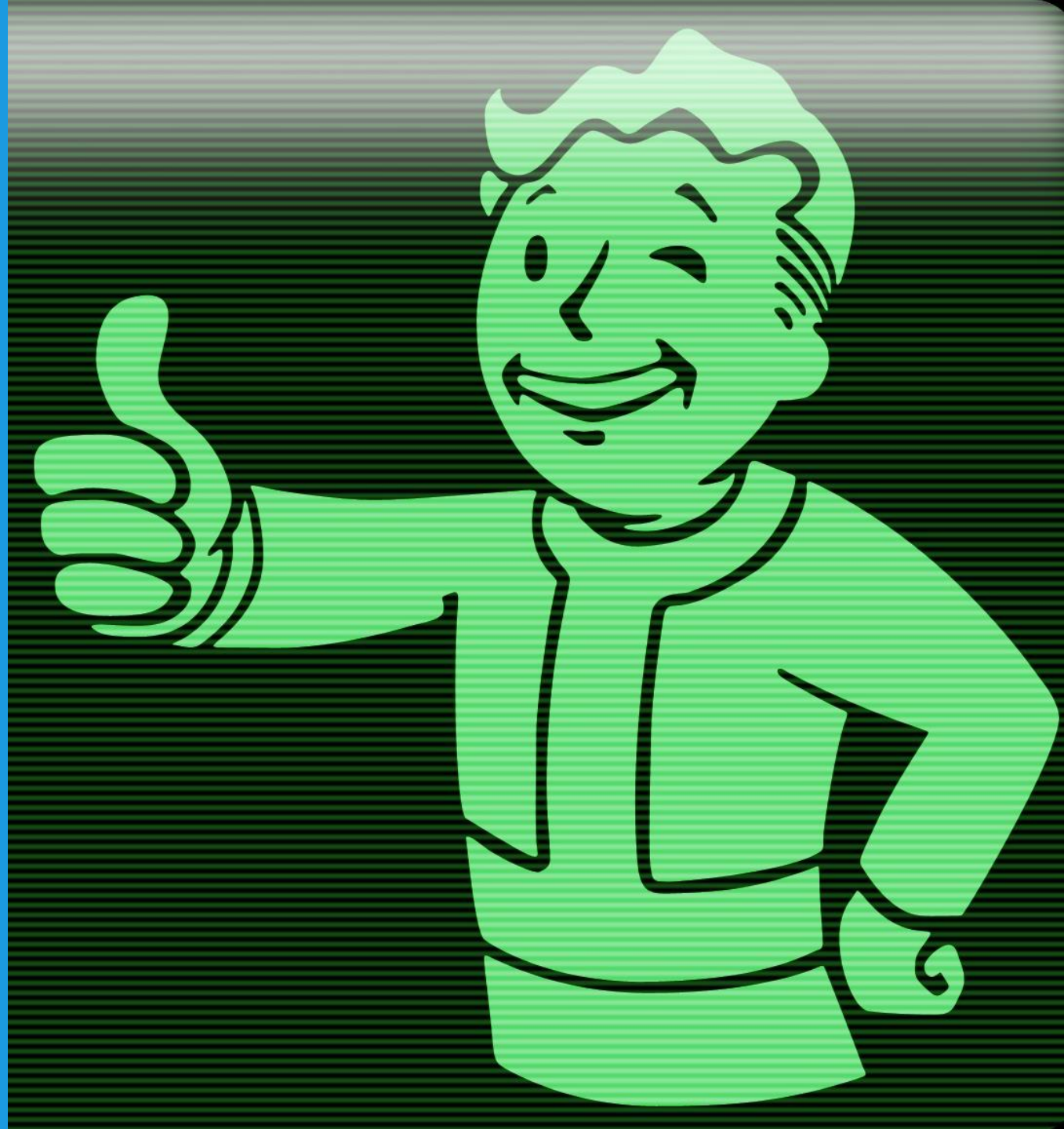
No changes were requested that day!



Questions?

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Thank You!

Reference: Immutable building blocks

Radius is an open-source, cloud-native, application platform from Microsoft that is contributed to the Cloud Native Compute Foundation.

It allows you to develop (platform agnostic) recipies that developers can use in their solutions.

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